



Mid-Semester Project Presentation

Agentic AI and RAG-Based Framework for Digital Forensics

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STATUS AT MID-SEMESTER

The framework runs end to end on local models, behind an API and a web interface.

6 of 7 sub-objectives complete; evaluation is complete for the local tools.

81 automated tests and CI pass.

Language-model judges are built but not yet benchmarked.

Introduction

THE PROBLEM

- Generative AI produces fluent text and photorealistic faces at very low cost.
- A single detector returns a probability but no evidence, and often fails on a new generator, domain or compression level.
- Forensic use needs the evidence examined, how reliable the result is, and a clear signal when a person must decide.

WHERE IT APPLIES

- Digital evidence screening in cybercrime investigations
- Newsroom and fact-checking workflows
- KYC and identity-fraud checks on facial media
- Enterprise media verification
- Academic and publishing screening of machine-written text

PROBLEM STATEMENT

Design an end-to-end framework that examines text and images with complementary tools, combines their outputs in a controlled way, cites retrieved supporting knowledge, and refers doubtful cases to a person.

Introduction: Concepts and Datasets

TECHNICAL CONCEPTS USED

- **Fine-tuned transformer:** DeBERTa-v3-base text classifier
- **Binoculars:** perplexity ratio of two language models
- **CNN with Grad-CAM:** EfficientNet-B4 face-swap detector
- **CLIP linear probes:** face, general-image and scope guard
- **Calibration:** temperature scaling on a separate split
- **RAG:** ChromaDB with a similarity cut-off and cited sources
- **Reflexion:** one judge re-check when tools disagree
- **LangGraph:** typed-state orchestration

DATASETS AND INPUT FORMAT

Modality	Data	Use
Text	HC3, MAGE, RAID	136k-row training mix; held-out sets
Face swap	WildDeepfake, FF++ (C32), Celeb-DF v2	Training mix; held-out tests
Synthetic faces	GRAVEX-200K, DeepDetect-2025	CLIP face probe
Art, scenes	DeepDetect, AI-art set	General image probe

Input: UTF-8 text (20 to 20,000 words) or JPEG, PNG, WebP up to 10 MB, checked by content.

Leakage control: text split by source document, images by video; Cohere and the paraphrase, synonym and homoglyph attacks never used in training; Face2Face and FaceShifter are the zero-shot test.

Literature Review: Related Work

Work	Area	Contribution	Limitation or gap
Dugan et al. (RAID), 2024	Text	Robustness benchmark: 11 generators, 8 domains, 11 attacks	In-domain accuracy alone is insufficient
Hans et al. (Binoculars), 2024	Text	Zero-shot detection with two language models	Varies with generator and domain
Rossler et al. (FF++), 2019	Image	Controlled facial-manipulation benchmark	Depends on method and compression
Zi et al. (WildDeepfake), 2020	Image	Real-world, in-the-wild deepfake data	Detection only, no evidence reasoning
Lewis et al. (RAG), 2020	RAG	Retrieval gives explicit external knowledge	Output quality follows retrieval quality
Zhang et al. (ForenAgent), 2026	Agentic	Iterative tool use for forgery analysis	Latency and reliability are open

Literature Review: Inference and Gap

INFERENCE FROM THE LITERATURE

- **Performance is distribution-dependent.** Unseen generators, attacks and compression lower accuracy.
- **Signals are complementary.** Statistical, learned and semantic evidence fail in different places.
- **Explanation must be tied to evidence.** Retrieval helps, but scores and uncertainty must stay visible.

RESEARCH GAP

No unified text-and-image workflow that combines tools, grounds the result in retrieved knowledge and verifies disagreement before reporting.

SEEN IN OUR OWN FIRST MODELS

0.66

text AUROC on MAGE, after 1.00 on HC3

0.52

image AUROC on FF++ (chance), after 0.97 on WildDeepfake

SWOT

Strengths	Weaknesses
Multi-signal, modular, explainable	Latency; calibration effort
Opportunities	Threats
Video, provenance, watermarking	New generators; evasion

MAIN OBJECTIVE

Develop an Agentic AI and RAG-based framework that detects AI-generated or manipulated text and images with complementary tools and produces explainable, evidence-grounded, confidence-aware reports.

No.	Sub-objective	Status	Basis
1	Multi-signal text detector	Done, modified	DeBERTa-v3, Binoculars, LLM judge
2	Image forensics pipeline	Done, modified	EfficientNet-B4, CLIP probes, scope guard
3	LangGraph orchestrator	Done	Routing, verifier loop, Reflexion
4	Shared RAG knowledge base	Done	Literature, generator profiles, case memory
5	Verifier and escalation	Done	Calibrated fusion, human referral
6	Evaluation and ablations	Partly done	9 held-out sets; judge rows pending
7	Prototype and deployment	Done, modified	FastAPI, web interface, Docker, CI

Methodology

REFERENCE MODEL

Iterative, evaluation-driven development with frozen thresholds and untouched test sets.

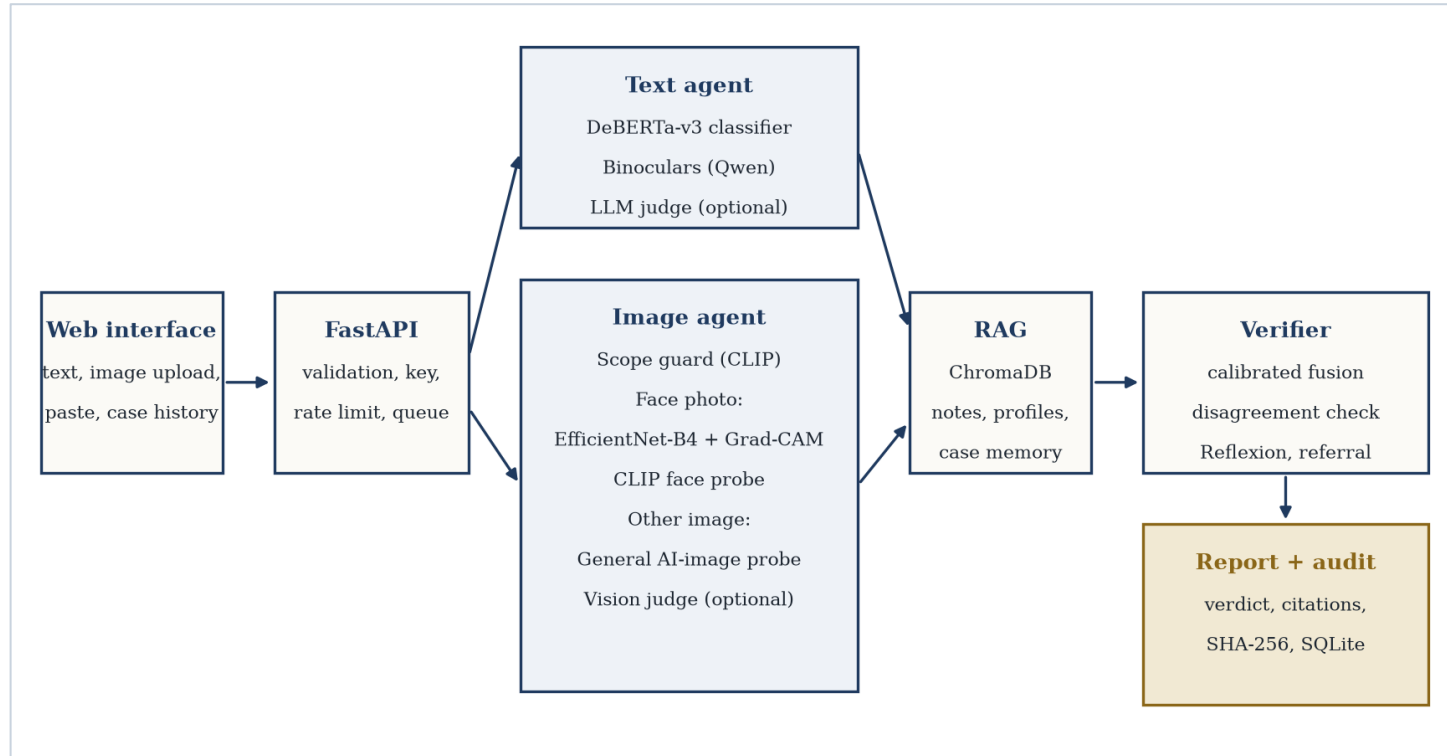
STEPS

- Prepare data with leakage checks
- Train each tool with early stopping
- Calibrate on a separate split
- Orchestrate, fuse and verify
- Serve through API and web interface
- Evaluate on held-out data; diagnose; fix

TIMELINE AND DELIVERABLES

Ph.	Task	Deliverable	Status
A	Literature, architecture	Gap analysis, state schema	Done
B	Text models	Classifier, probe, judge	Done
C	Image models	CNN, CLIP, scope guard	Done
D	Orchestration	Routing, fusion, Reflexion	Done
E	RAG and verifier	Knowledge base, citations	Done
F	Interface, evidence store	Web UI, audit records	Done
G	Evaluation, report	Ablations, report	In progress

Working Model: Architecture



REQUIREMENTS

Functional

- Accept text or image
- Verdict with confidence
- Per-tool readings and Grad-CAM
- Cited reference material
- Referral of doubtful cases
- Hashed audit record

Non-functional

- Runs on one 16 GB GPU
- Inputs validated by content
- No secrets in the repository
- Reproducible tests and CI

Working Model: Modules and Deliverables

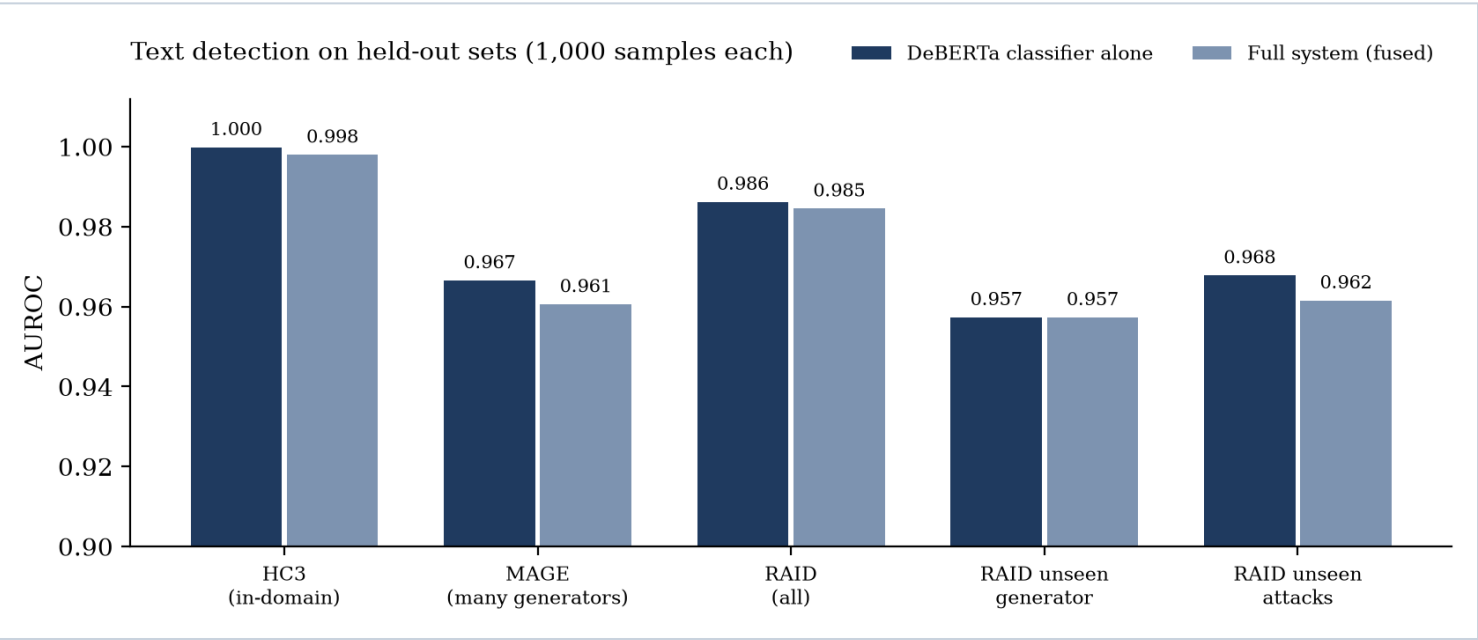
Module	Attained	Remaining
Core engineering	Typed state, settings, JSON logs, 81 tests, CI with Docker build	Distributed tracing
Text agent	DeBERTa-v3, Binoculars, judge adapter, input normalisation, calibration	Judge calibration
Image agent	Scope guard, EfficientNet-B4, CLIP probes, Grad-CAM, robustness test	Unseen-method transfer
Orchestration	Routing, fusion, disagreement rule, Reflexion, referral	Live Reflexion measurements
RAG	21-chunk knowledge base, cut-off and query rewrite, case memory with corrections	Larger curated corpus
Product	Web interface, API key, rate limit, SQLite audit store, Docker	PDF export of a case

Changes from the synopsis, with reasons, are kept in DECISIONS.md: XGBoost removed, DeBERTa-v3 instead of RoBERTa, CLIP probes instead of frequency tools, web interface instead of React.



- **Finding and meter:** verdict, confidence and the combined reading against the decision thresholds
- **Instrument readings:** each tool's score and reliability, or why it was not used
- **Grad-CAM figure:** regions that influenced the network
- **Verifier notes and referral:** why a case was or was not sent to a person
- **Reference material:** cited sources; the system's own earlier cases are marked unverified
- **Examiner's determination:** a human label that feeds case memory

Results: Text



Input normalisation alone lifted the homoglyph attack from AUROC 0.72 to 0.99, without retraining. Binoculars adds no accuracy (fused AUROC -0.001 to -0.006), so its weight is capped at 0.10.

0.957 to 0.998

AUROC on five held-out sets, including an unseen generator and unseen attacks

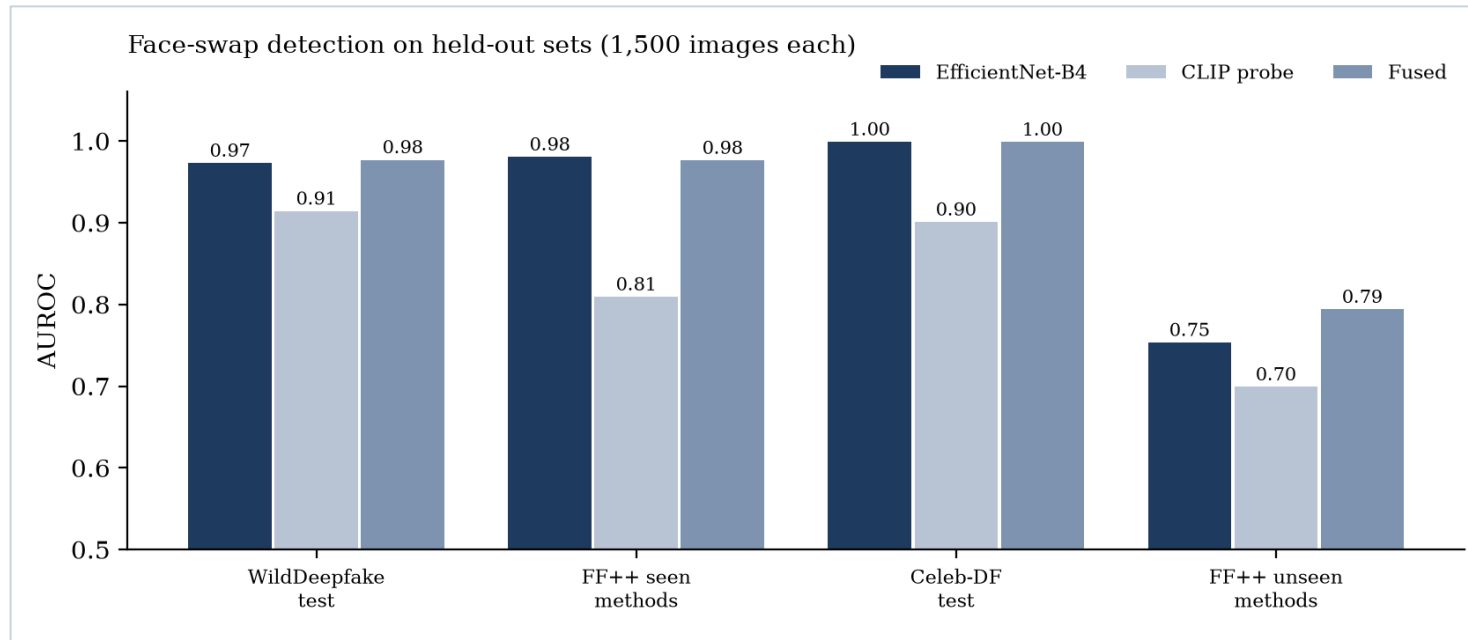
58 to 61%

AI text caught at 1% false positives on the hardest sets (MAGE, unseen attacks)

92 to 99.6%

accuracy on cases the system decides; 30 to 52% are referred to a person

Results: Images



Self-blended images, one extra manipulation method and a CLIP-only probe each failed to close the unseen-method gap; fusing CLIP with the CNN gave +0.04.

0.976 to 0.999

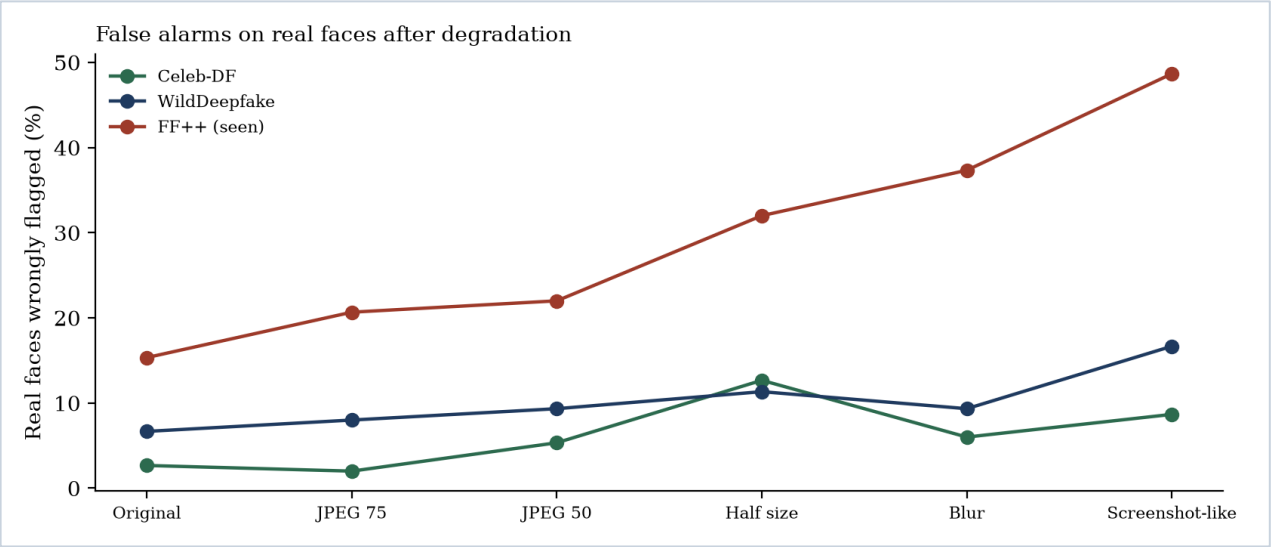
AUROC on WildDeepfake, FF++ (seen) and Celeb-DF

0.794

AUROC on unseen methods; 13% of fakes caught at 1% false positives

A model trained on WildDeepfake alone scored 0.52 on FF++. Cause: full frames versus face crops, and one training source. Face crops plus a three-dataset mix raised it to 0.98.

Results: Robustness and Fixes



Real faces wrongly flagged rise from 3% to 9% (Celeb-DF), 7% to 17% (WildDeepfake) and 15% to 49% (FF++) under screenshot-like damage, while AUROC mostly holds.

A second threshold fitted on degraded calibration images halves false alarms on Celeb-DF (9% to 4%) and WildDeepfake (17% to 8%) for 2 to 4 points of fakes caught. FF++ stays at 37%.

Problem found	Fix and effect
Text 0.66 on MAGE	Mixed corpora, held-out generators: 0.97
Images 0.52 on FF++	Face crops, three sources: 0.98
Homoglyph attack 0.72	Input normalisation: 0.99
Illustration judged authentic	Scope guard, general probe
75% of MAGE referred	Weak tools advisory: 52%
Stale image calibration	Re-scored, refit, benchmarks rerun

Conclusion

JUSTIFICATION OF OBJECTIVES

- End-to-end system built and tested: 81 tests, CI and Docker build green
- Calibrated fusion with referral: 92 to 99.6% accuracy on decided cases (text and seen image sets)
- Cited, auditable reports with human corrections feeding case memory

PLAN FOR THE END-SEMESTER STAGE

- Calibrate and benchmark the judges; measure Reflexion
- Detect re-uploads and apply the degraded-image threshold
- Second unseen synthetic-face set; non-native-English text audit
- Retrain the CNN with synthetic faces; test frequency evidence
- PDF export of a case; final report

LIMITATIONS WE STATE OPENLY

Unseen deepfake methods reach only 0.794 AUROC. Re-uploaded images raise false alarms. The vision judge missed the demo deepfake (0.05 with 85% confidence), so judge weights are kept low until calibrated. Detection is probabilistic, not proof.

Reference

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- [7] Zhang et al., Code-in-the-Loop Forensics, arXiv:2512.16300, 2026
- [8] Wang et al., LaP-Forensics, arXiv:2607.25962, 2026

B: ADDED DURING IMPLEMENTATION

- [9] Selvaraju et al., Grad-CAM, ICCV 2017
- [10] Tan and Le, EfficientNet, ICML 2019
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- [12] He et al., DeBERTa, ICLR 2021
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- [14] Shinn et al., Reflexion, NeurIPS 2023
- [15] Li et al., MAGE: Machine-generated Text Detection in the Wild, ACL 2024
- [16] Yan et al., Corrective Retrieval Augmented Generation, arXiv:2401.15884



Thank You